MODULE 1 · LECTURE 32

Case Studies and Industry Trends in Agricultural Informatics and AI

How to read an agri-tech case study critically, five Indian case studies — AI sowing advice, Saagu Baagu, AI pest traps, drones run by women's self-help groups and AI advisory services — the industry trends shaping the sector, a business-case analysis of a spraying drone, and a review of the whole course.

Learning objectives

After this lecture you should be able to:

1. Apply a framework to judge the claims and evidence of an agri-tech case study.
2. Describe Indian case studies of AI and digital technology in agriculture and their lessons.
3. Identify current industry trends: agri-startups, drones as a service, AI advisory and automation.
4. Evaluate a technology investment with net present value, payback and Monte Carlo simulation.
5. Explain the barriers to inclusive adoption of digital automation.
6. Connect the topics of the course and identify skills needed by agricultural engineers.

Scope: this final lecture brings together the course. Earlier case studies of ICT — Kerala fisheries, e-Choupal, Bhoomi and Digital Green — were covered in Lecture 10; this lecture focuses on AI-era cases and on where the industry is heading.

1. Reading a case study critically

Agri-tech success stories are often reported by the organisations that run them. Evidence on the impact of ICT services for farmers is mixed, and impact depends on context, user experience and trust (Aker et al., 2016). Before accepting a result, ask:

Question	What to look for
What problem was solved?	A real constraint for farmers — not just a new technology looking for a use
What did the technology do?	Sensing, data, AI model, decision, action, delivery channel (SMS, app, service provider)
Who reported the result?	Independent study, peer-reviewed paper, government report or the implementing partner
How was the effect measured?	Comparison group, number of farmers, seasons, crops; one good season can mislead
Who paid and who benefits?	Business model: fee, subsidy, advertising, data; is it sustainable after the pilot?
Can it scale?	Connectivity, language, cost, skills, service network, institutions

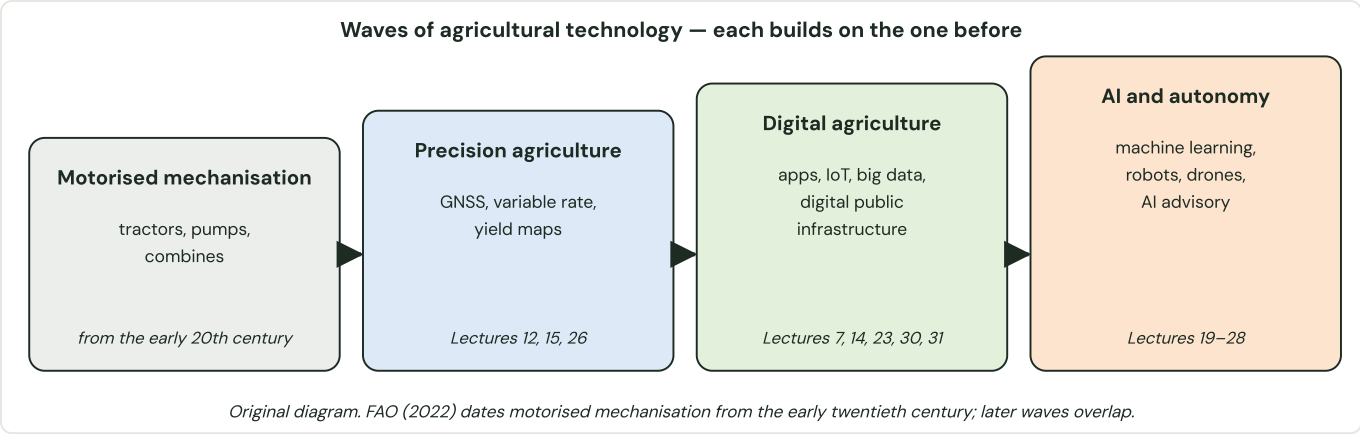


Figure 1. Waves of agricultural technology. Original diagram.

2. Indian case studies

2.1 AI sowing advisory for groundnut (Andhra Pradesh)

In June 2016 ICRISAT, Microsoft and the Government of Andhra Pradesh piloted a **Sowing App** and a personalised village advisory dashboard for groundnut farmers in Devanakonda mandal, Kurnool district. Using machine-learning tools, weather forecasts and data including 45 years of rainfall and 10 years of groundnut sowing progress, it advised the best sowing week. Registered farmers received 10 SMS advisories in Telugu from 10 June on sowing, seed treatment, sowing depth, weed management, manure, harvesting and storage. Farmers who sowed in the first week of June suffered from a long dry spell in August, while registered farmers who sowed in the last week of June and first week of July as advised obtained better yields; the partners reported a **30% higher average yield per hectare** (Microsoft, 2017).

- **Lesson**: AI need not be complex — the right advice on one decision (when to sow), delivered by SMS in the local language, can matter most in a drought year.
- **Caution**: one crop, one season, a pilot area, and a comparison with farmers who sowed early, as reported by the partners.

2.2 Saagu Baagu — AI for chilli farmers (Telangana)

Under the World Economic Forum's Artificial Intelligence for Agriculture Innovation (AI4AI) initiative, the Government of Telangana piloted **Saagu Baagu** in Khammam district, supported by the Bill & Melinda Gates Foundation and implemented by Digital Green, for more than 7,000 chilli farmers. The World Economic Forum (2024) reports that in the first phase farmers saw a **21% increase in chilli yield per acre**, 9% less pesticide use, 5% less fertiliser use and 8% better unit prices from improved quality, raising incomes by more than ₹66,000 per acre per crop cycle. In October 2023 the state expanded the programme to potentially reach 5,00,000 farmers across five value chains.

- **Lesson**: a bundle of agritech and data services built around one value chain, with the state government as anchor, can scale quickly.
- **Caution**: results are programme-reported; independent evaluation over several seasons is needed. For a 2-acre chilli farm the reported gain would be 2 × ₹66,000 = ₹1.32 lakh per crop cycle — large enough that verification matters.

2.3 AI pheromone traps for pink bollworm

ICAR-CICR piloted camera-equipped pheromone traps with 18 cotton farmers in Punjab: a machine-learning model identifies and counts pink bollworm moths in trap images and farmers receive updates

on their phones (Shagun, 2024; Lecture 25). **Lesson**: AI makes the economic threshold usable by automating the tedious step of counting.

2.4 Namo Drone Didi — drones as a service by women's self-help groups

The Central Sector Scheme **Namo Drone Didi**, with an outlay of ₹1,261 crore for 2023-24 to 2025-26, aims to provide 15,000 drones to women self-help groups (SHGs) so that they can earn income as drone service providers. It gives 80% of the cost of the drone package, up to ₹8 lakh, with 15 days of drone-pilot training for one member and 5 days of drone-assistant training for another. A study of 500 scheme drones by ADRTC Bangalore found that a drone covers one acre in 7–8 minutes and flies 5–20 minutes on one battery charge; where no utility vehicle was provided, 42.68% of drone didis faced transport problems, so assistance for multi-utility vehicles was added (Press Information Bureau [PIB], 2025a). By January 2026, lead fertiliser companies had distributed 1,094 drones to SHGs, of which 500 were under the scheme (PIB, 2026a).

- **Lesson**: new technology reaches small farms faster as a **service** run by local entrepreneurs than as something each farmer buys.
- **Caution**: 1,094 drones distributed against a target of 15,000 is about 7.3%; logistics such as transport can decide success.

2.5 AI advisory services built on digital public infrastructure

Kisan e-Mitra, a voice-enabled AI chatbot in 11 languages, had answered more than 93 lakh scheme queries by December 2025, and AI-based local monsoon-onset forecasts reached 3.88 crore farmers by SMS in a kharif 2025 pilot (PIB, 2026b; Lecture 31). **Lesson**: national platforms can bring AI to crores of farmers at very low cost per farmer.

Case	Technology	Evidence reported	Key lesson
AI sowing advisory (2016)	ML + weather data, SMS	30% higher average groundnut yield (partners)	One well-timed decision matters
Saagu Baagu (phase 1)	AI advisory + services bundle	+21% yield, -9% pesticide (WEF)	Value-chain bundle, state anchor
AI pheromone traps	Camera + ML counting	Pilot with 18 farmers	Automate tedious monitoring
Namo Drone Didi	Drones as a service	1,094 drones; 42.68% faced transport issues	Service model; logistics
Kisan e-Mitra, AI forecasts	Chatbot, AI weather	93 lakh queries; 3.88 crore farmers	Scale through public platforms

3. Industry trends

- **Agri-startups**: under the RKVY "Innovation and Agri-Entrepreneurship Development" programme (from 2018-19), six knowledge partners and 24 agribusiness incubators have trained more than 6,000 agri-startups and supported 2,096 of them during 2019-20 to 2025-26 with ₹168.14 crore of grants; selected start-ups can receive up to ₹5 lakh at the idea/pre-seed stage and ₹25 lakh at the seed stage. They work in precision agriculture (sensors, AI, IoT, ICT and drones), farm mechanisation, post-harvest, food technology, supply chains and waste-to-wealth (PIB, 2025b).
- **Digital automation**: a new generation of digital automation technologies can raise productivity and resilience and address environmental problems of past mechanisation. FAO's analysis of 27 case studies found key barriers to inclusive adoption by small-scale producers: low digital literacy, lack of

connectivity and electricity, and financial constraints (Food and Agriculture Organization of the United Nations [FAO], 2022).

- **AI advisory in Indian languages**: chatbots and voice services, and the proposed Bharat-VISTAAR tool linking AgriStack portals with ICAR advice (PIB, 2026b; Lecture 31).
- **Services instead of ownership**: drones, robots and sensors offered through custom-hiring centres, FPOs and SHGs (Lectures 24, 27).
- **Less chemical, more precision**: spot spraying and robotic weeding (Lectures 25, 27), where reported herbicide savings exceed 50% for patch spraying (Gerhards et al., 2022).
- **Robots still maturing**: harvesting robots reviewed up to 2014 averaged 66% harvest success and none was commercialised (Bac et al., 2014); progress since then is real but gradual (Lecture 27).

4. Worked example: is a spraying drone a good business?

An SHG considering a spraying drone must ask whether its income will repay the investment. **Net present value (NPV)** discounts future cash flows to today's value; a project with NPV above zero earns more than the discount rate.

$$\text{NPV} = -\text{Investment} + \Sigma\ \text{Cash}_t \div (1+\text{r})^t \qquad \text{Annuity factor (5 years, r} = 10\%) = 3.791 \qquad \text{Payback} = \text{Investment} \div \text{yearly cash}$$

Worked example 1 — a typical year
Assume 75 spraying days, 25 acres a day, a service charge of ₹375 and a cost of ₹155 per acre, and ₹60,000 of fixed costs a year (illustrative values).
Spraying 25 acres at 7–8 minutes an acre takes about 25 × 7.5 ÷ 60 ≈ 3.1 hours of flying a day (PIB, 2025a) — a realistic day with battery changes.
Yearly cash = 75 × 25 × (375 - 155) - 60,000 = **₹3,52,500**.
Without subsidy (₹10 lakh): NPV = -10,00,000 + 3,52,500 × 3.791 = **₹3.36 lakh**; payback = 10,00,000 ÷ 3,52,500 = 2.8 years.
With the 80% subsidy (₹2 lakh to pay): NPV = **₹11.36 lakh**; payback = 0.57 years.

But demand, prices and costs are uncertain. The program below draws 10,000 possible futures — spraying days, acres per day, fees and costs chosen at random within plausible ranges — and calculates the NPV of each, with and without the subsidy.

Monte Carlo business case for a spraying drone in Python (Lecture 8)

```
import random

random.seed(32)
PACKAGE = 10_00_000                  # drone package cost, Rs (illustrative)
SUBSIDY = min(0.8 * PACKAGE, 8_00_000)   # Namo Drone Didi: 80%, up to Rs 8 lakh
YEARS, RATE = 5, 0.10                # working life and discount rate
FIXED = 60_000                       # insurance, repairs, battery wear per year, Rs (illustrative)
ANNUITY = sum(1 / (1 + RATE) ** t for t in range(1, YEARS + 1))

def yearly_cash(days, acres_per_day, fee, cost_per_acre):
    return days * acres_per_day * (fee - cost_per_acre) - FIXED

def npv(investment, cash):
    return -investment + cash * ANNUITY

# Monte Carlo: demand and prices are uncertain, so try 10,000 possible futures
runs = []
for _ in range(10_000):
    days = random.uniform(30, 100)           # spraying days a year (seasonal demand)
    acres = random.uniform(15, 35)           # acres sprayed per working day
    fee = random.uniform(300, 450)           # service charge, Rs per acre
    cost = random.uniform(130, 180)          # pilot's wage, charging, transport, Rs per acre
    runs.append(yearly_cash(days, acres, fee, cost))

for label, invest in (("without subsidy", PACKAGE), ("with subsidy", PACKAGE - SUBSIDY)):
    values = sorted(npv(invest, c) for c in runs)
    p_gain = 100 * sum(v > 0 for v in values) / len(values)
    print(f"{label:16s} investment Rs {invest / 1e5:5.2f} lakh   median NPV Rs {values[5000] / 1e5:5.2f}
lakh"
          f"   chance NPV > 0: {p_gain:5.1f}%")

print("Break-even spraying days a year (25 acres/day, fee Rs 375, cost Rs 155/acre):")
for label, invest in (("without subsidy", PACKAGE), ("with subsidy", PACKAGE - SUBSIDY)):
    days = (invest / ANNUITY + FIXED) / (25 * (375 - 155))
    print(f"  {label:16s} {days:5.1f} days")

Output:
without subsidy  investment Rs 10.00 lakh   median NPV Rs  0.38 lakh   chance NPV > 0:  52.4%
with subsidy     investment Rs  2.00 lakh   median NPV Rs  8.38 lakh   chance NPV > 0:  98.4%
Break-even spraying days a year (25 acres/day, fee Rs 375, cost Rs 155/acre):
  without subsidy   58.9 days
  with subsidy      20.5 days
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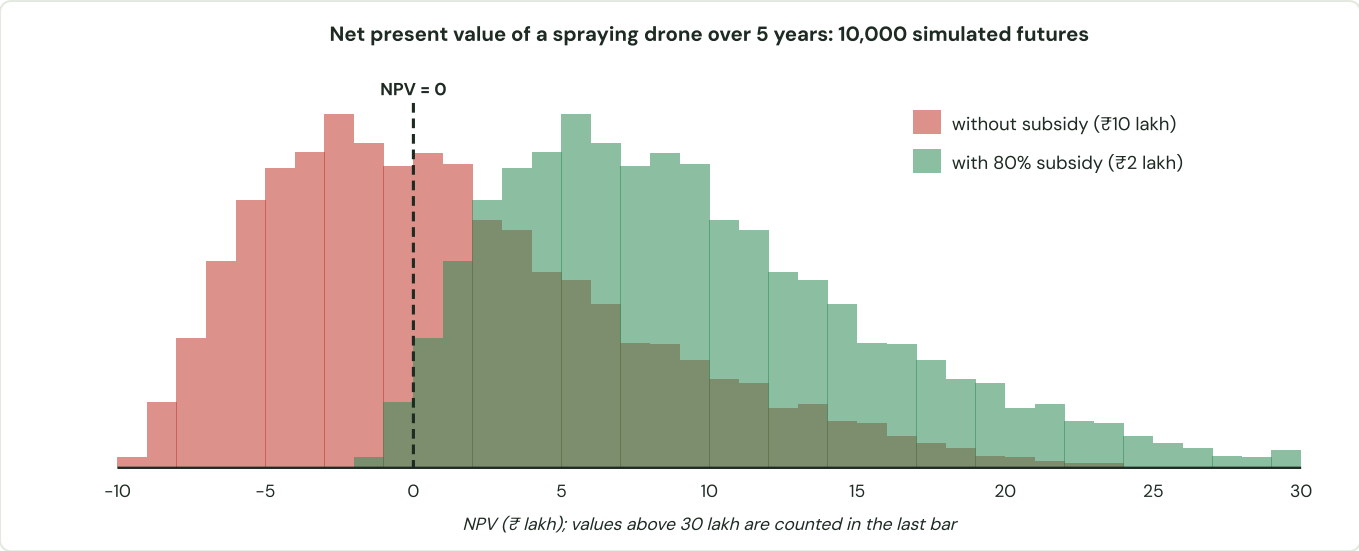


Figure 2. Distribution of net present value over 10,000 simulated futures, with and without the subsidy. Original chart from the program's data.

- Without the subsidy, the drone pays off in only about half of the futures (52.4%) — a risky investment for a self-help group.
- With the 80% subsidy, it pays off in 98.4% of futures, with a median NPV of ₹8.38 lakh.
- The break-even demand falls from about 59 to about 21 spraying days a year: the subsidy mainly protects the group against poor seasons.
- Demand (spraying days) is the biggest risk — which is why transport, bookings and links with FPOs and fertiliser companies matter as much as the drone itself.

The program was run and the output shown is its actual output. The package cost, fees, costs and ranges are illustrative assumptions; the subsidy rule, coverage time and transport finding are from PIB (2025a).

5. Skills for the future

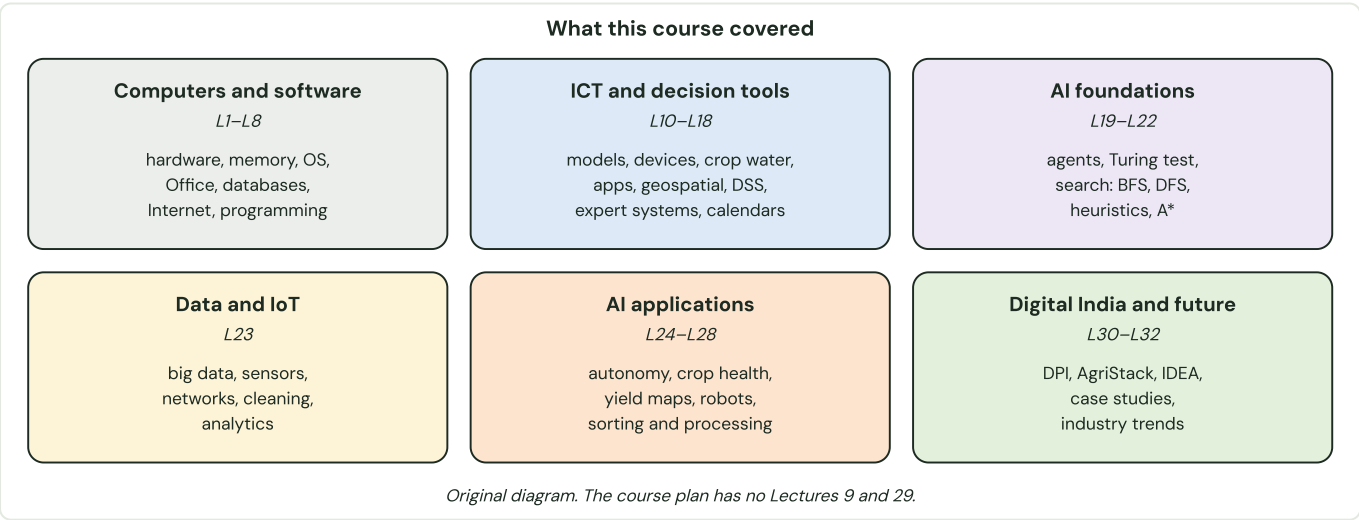


Figure 3. The six themes of this course. Original diagram.

The cases in this lecture combine almost every topic of the course. An agricultural engineer working in this field needs to be able to:

- understand crops, soils, machines and farmers' decisions — the problem comes first;
- work with sensors, devices and data (Lectures 2, 12, 23) and databases (Lecture 6);
- program and analyse data (Lecture 8) and use models and decision tools (Lectures 11, 16, 17);

- use geospatial data and precision equipment (Lectures 15, 26);
- build, evaluate and explain AI models honestly — accuracy, precision and recall, lab versus field (Lectures 19–28);
- judge the economics of a technology, as in Section 4;
- respect privacy, consent and inclusion (Lectures 30, 31).

6. Key takeaways

- Judge agri-tech claims by asking who reported them, how they were measured, who pays and whether they can scale.
- AI sowing advice (30% higher groundnut yield in a dry year) and Saagu Baagu (+21% chilli yield, programme-reported) show the value of timely, local advice.
- AI pest traps automate monitoring; Namo Drone Didi delivers drones as a service through women's SHGs, with logistics as a key constraint.
- National platforms such as Kisan e-Mitra bring AI to crores of farmers.
- Trends: agri-startups (2,096 supported), digital automation, AI advisory in Indian languages, services instead of ownership, less chemical use.
- FAO identifies low digital literacy, poor connectivity and electricity, and finance as the main barriers for small producers.
- NPV and Monte Carlo simulation show how subsidies and demand decide whether a technology pays.

7. Examination questions

Short-answer questions (2 marks)

Q1. Give four questions to ask when reading an agri-tech case study.

Any four: what problem, what technology, who reported, how measured, who pays, can it scale.

Q2. What did the ICRISAT–Microsoft Sowing App advise farmers?

The best time (week) to sow groundnut, with SMS advisories in Telugu on sowing and crop management.

Q3. What results were reported for Saagu Baagu's first phase?

21% higher chilli yield per acre, 9% less pesticide, 5% less fertiliser and 8% better prices (WEF, 2024).

Q4. What is the Namo Drone Didi scheme?

A central scheme to provide 15,000 drones to women SHGs as drone service providers, with 80% assistance up to ₹8 lakh.

Q5. How long does a kisan drone take to spray one acre, according to the ADRTC study?

7–8 minutes.

Q6. How many agri-startups were supported under the RKVY programme during 2019-20 to 2025-26?

2,096, with ₹168.14 crore of grants.

Q7. Name three barriers to inclusive adoption of digital automation identified by FAO (2022).

Low digital literacy, lack of connectivity and electricity, and financial constraints.

Q8. Define net present value.

The sum of future cash flows discounted to today, minus the investment.

Q9. What is a Monte Carlo simulation?

Repeating a calculation many times with randomly drawn uncertain inputs to see the range of outcomes.

Q10. Why did the drone subsidy matter most in poor seasons?

It lowered the break-even demand from about 59 to about 21 spraying days a year.

Long-answer questions (5–10 marks) — answer outlines

Q1. How should an agri-tech case study be evaluated? Illustrate with one Indian example.

Framework table; Aker et al. (2016); apply to the sowing app or Saagu Baagu, with cautions.

Q2. Describe two Indian case studies of AI in agriculture and the lessons they offer.

Sowing app; Saagu Baagu; AI traps; Kisan e-Mitra; summary table.

Q3. Discuss drones as a service in Indian agriculture with reference to Namo Drone Didi.

Scheme features; training; ADRTC findings; distribution progress; worked example 1.

Q4. Discuss the major industry trends in agricultural informatics and AI.

Agri-startups; digital automation (FAO, 2022); AI advisory; services model; precision spraying; robots.

Q5. Evaluate the business case for a spraying drone using NPV and Monte Carlo simulation.

Formula; worked example 1; program; Figure 2; subsidy and demand risk.

Q6. Summarise the knowledge and skills an agricultural engineer needs for digital agriculture.

Figure 3; skills list; ethics, privacy and inclusion.

8. Glossary

Term	Meaning
Agribusiness incubator	Organisation that nurtures start-ups with training, mentoring and funds
Annuity factor	Present value of ₹1 a year for n years at rate r
Break-even	Level of activity at which NPV becomes zero
Case study	Detailed account of one project or programme
Comparison group	Similar farmers who did not use the technology
Discount rate	Rate used to convert future money to present value
Drone as a service	Spraying or imaging offered for a fee instead of owning a drone
Drone didi	Woman SHG member trained as a drone pilot
Monte Carlo simulation	Repeated calculation with random uncertain inputs
NPV	Net present value
Payback period	Years needed to recover the investment
Pilot (project)	Small first trial before scaling up
Programme-reported result	Result reported by the organisation running the programme
Scaling up	Extending a successful pilot to many more users
SHG	Self-help group
Value chain	Steps from input supply and production to the consumer

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All diagrams, charts, examples and programs are original. Case-study results are as reported by the cited organisations; most are not independent evaluations. Costs and prices in the drone example are illustrative.

Practice MCQs

Q1. When judging an agri-tech success story, the first thing to check is

(a) the colour of the app
(b) who reported the result and how it was measured
(c) the number of downloads only
(d) the company logo

Q2. The ICRISAT–Microsoft Sowing App pilot (2016) advised groundnut farmers mainly on

(a) fertiliser brand
(b) the best time to sow
(c) tractor hire
(d) export markets

Q3. Farmers following the sowing advisories reported a yield per hectare that was on average

(a) 10% lower
(b) the same
(c) 30% higher
(d) 300% higher

Q4. Saagu Baagu was piloted for chilli farmers in

(a) Kurnool, Andhra Pradesh
(b) Khammam, Telangana
(c) Nagpur, Maharashtra
(d) Punjab

Q5. The World Economic Forum reported Saagu Baagu phase 1 chilli yields per acre rising by

(a) 5% (b) 9%
(c) 21% (d) 66%

Q6. Namo Drone Didi provides drones to

(a) large companies (b) women self-help groups
(c) only research institutes (d) banks

Q7. Central assistance under Namo Drone Didi is

(a) 20% up to ₹2 lakh (b) 50% up to ₹5 lakh
(c) 80% up to ₹8 lakh (d) 100% without limit

Q8. In the ADRTC study, the share of drone didis facing transport problems where no utility vehicle was provided was

(a) 4.3% (b) 18%
(c) 42.68% (d) 90%

Q9. Under the RKVY agri-startup programme, the maximum seed-stage assistance is

(a) ₹5 lakh (b) ₹25 lakh
(c) ₹1 crore (d) ₹168 crore

Q10. FAO (2022) identifies which of these as a barrier to inclusive adoption of digital automation?

(a) too much electricity (b) low digital literacy
(c) excess connectivity (d) too many skills

Q11. A project is financially worthwhile when its net present value is

(a) negative (b) zero or above
(c) equal to the investment (d) unknown

Q12. In the drone simulation, the subsidy raised the chance of a positive NPV from about 52% to about

(a) 60% (b) 75%
(c) 98% (d) 100%

Answers

Q1 — (b) who reported the result and how it was measured
Q2 — (b) the best time to sow
Q3 — (c) 30% higher
Q4 — (b) Khammam, Telangana
Q5 — (c) 21%
Q6 — (b) women self-help groups
Q7 — (c) 80% up to ₹8 lakh
Q8 — (c) 42.68%
Q9 — (b) ₹25 lakh
Q10 — (b) low digital literacy
Q11 — (b) zero or above
Q12 — (c) 98%